

La_Fontaine_Fall_2025_Report2

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2025-11-25

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1 Summary

A complete randomized design that evaluates the effect between three treatments BFF (Fine flour), BWF (Whole flour) and BWG was conducted in the years 2024 and 2025. The response variables were APC, EB, Coliform, Yeast, Molds. There were in total 98 observations with only 26 of the observations from the year 2025. Data on the two years were analyzed separately because of some differences in the characteristics of the data. The client is interested in finding if there is any significant difference in effect between BWF and BWG, and BWG and BFF.

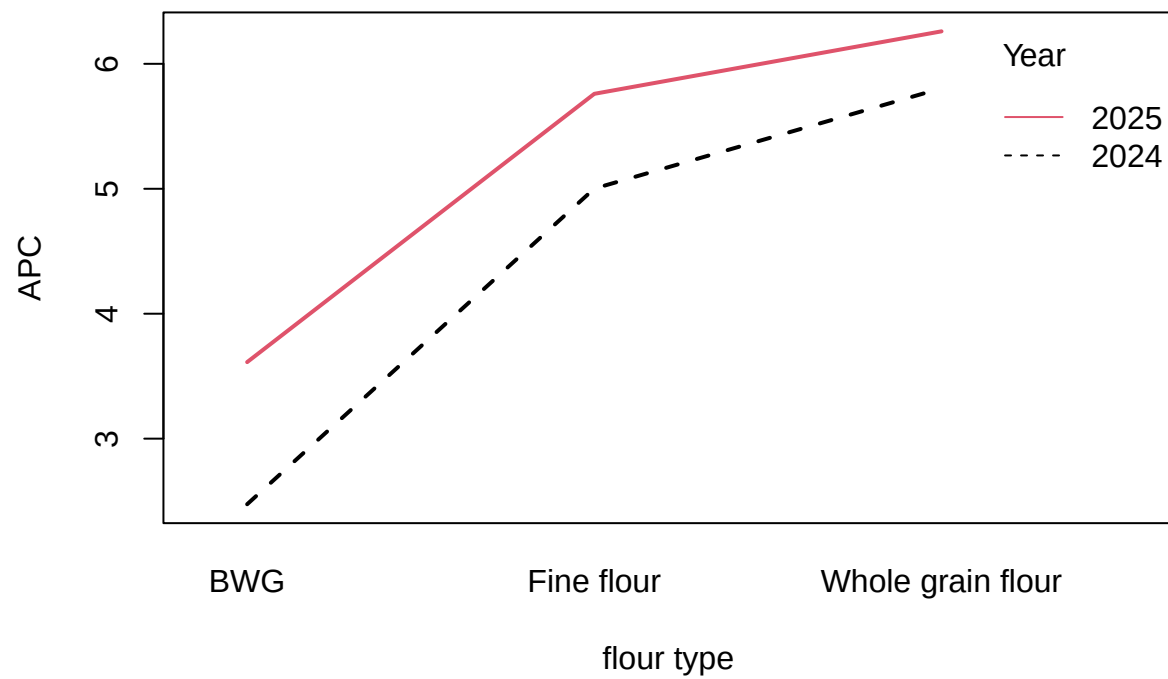
2 Data Description

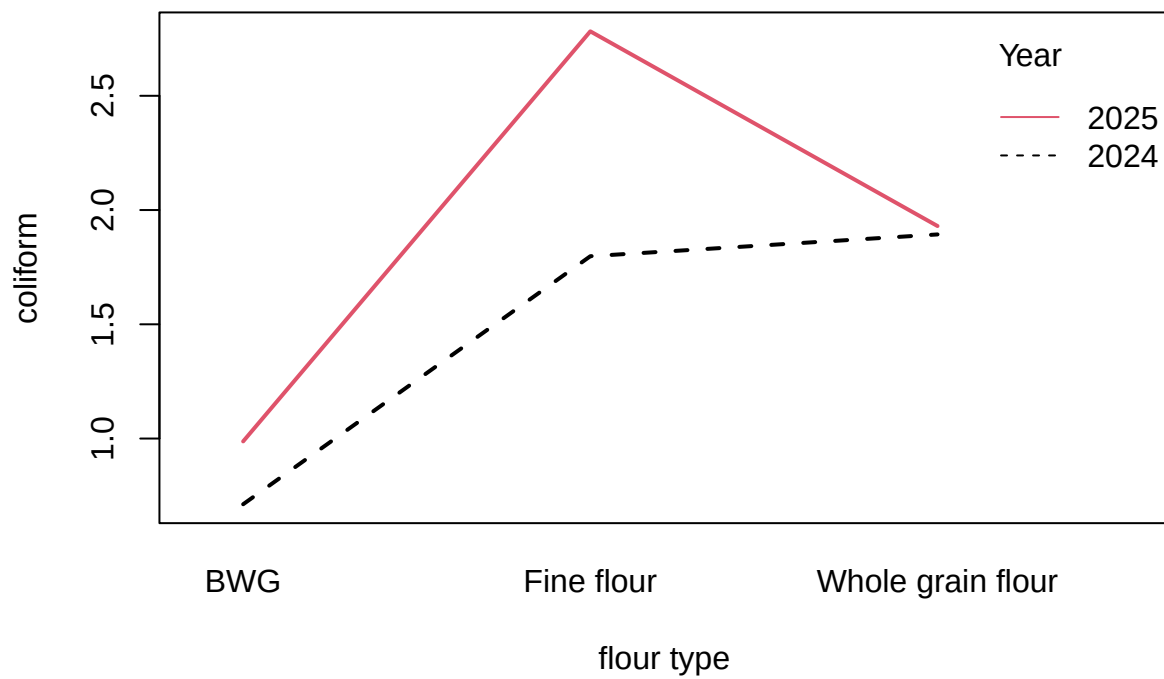
Data set contains seven variables, year (block), treatment and 5 response variables which will be analyzed separately. There were two blocks and three levels of treatment. The exploratory data analysis was done in R while the model fit was done using SAS.

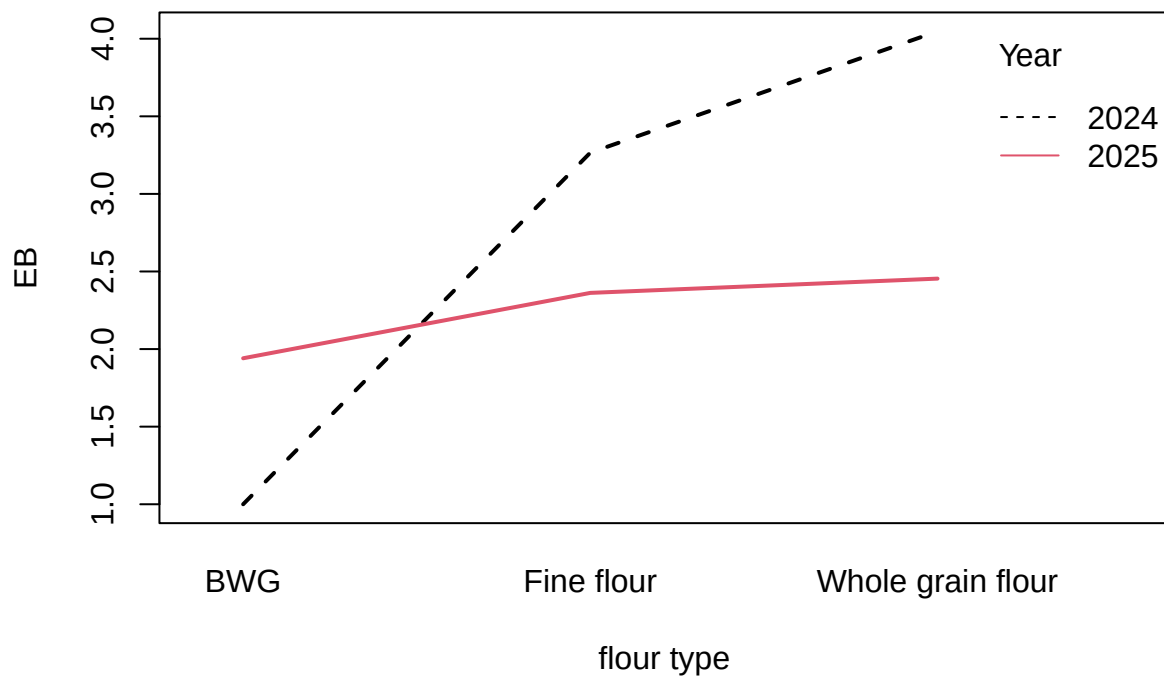
##		Flour_Type	Year	APC	coliform	EB	Yeast	Molds
## 1	Whole	grain flour	2024	5.73	3.60	4.45	2.56	2.59
## 2	Whole	grain flour	2024	6.26	3.78	4.64	4.02	3.92
## 3	Whole	grain flour	2024	5.69	0.50	3.00	3.93	3.08
## 4	Whole	grain flour	2024	3.97	0.50	1.78	1.70	1.78
## 5	Whole	grain flour	2024	6.30	2.48	4.28	3.53	3.18
## 6	Whole	grain flour	2024	5.98	2.70	3.60	3.49	3.83

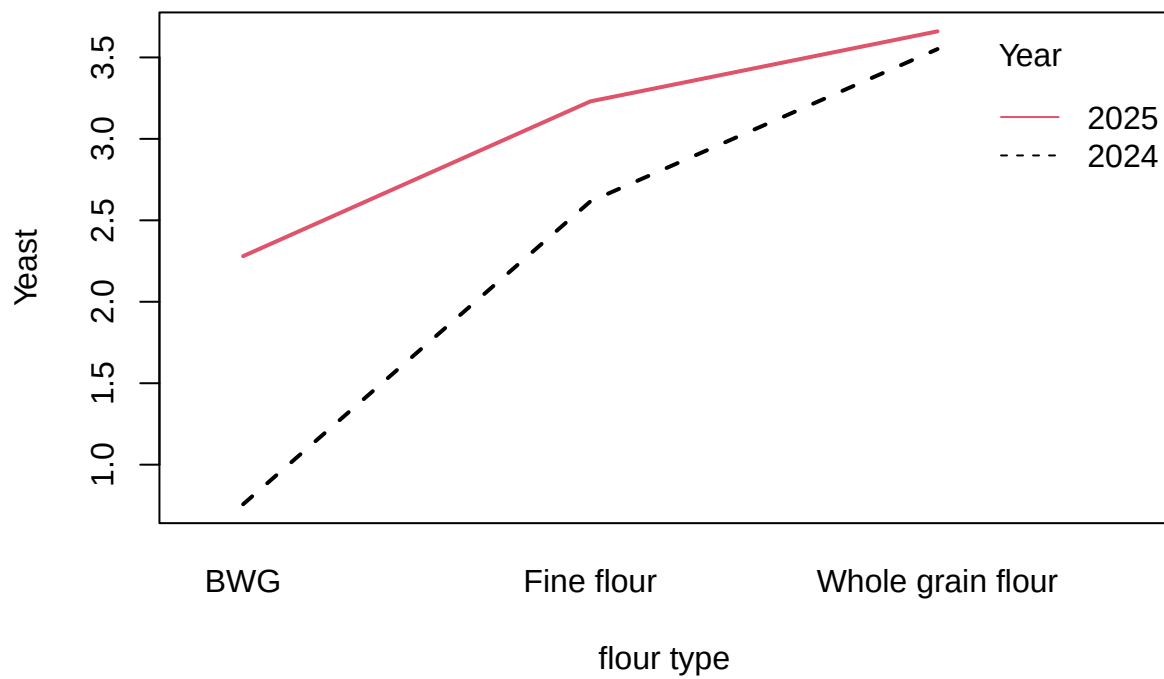
3 Exploratory Data Analysis (EDA)

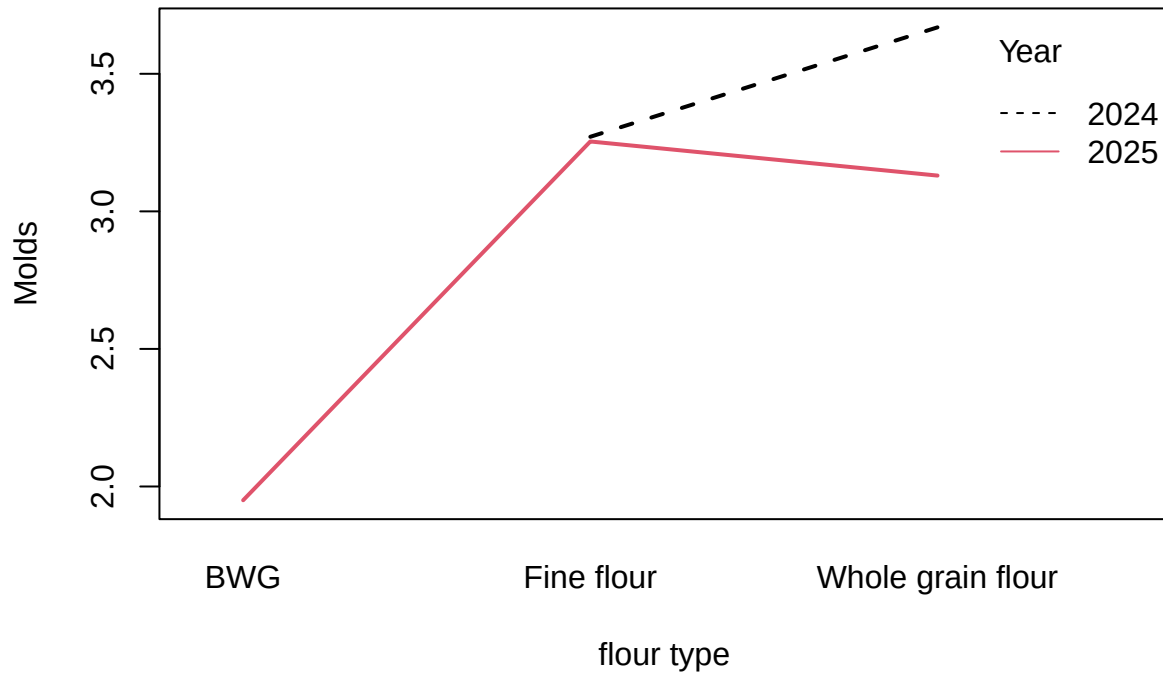
Before fitting a formal statistical model, we explore the data visually to identify trends, patterns, and potential interactions between the factors. We show interaction plots for each of the response variables to determine if there is any significant block by treatment interaction.











4 Interpretation of interaction plot

The plots above show the interaction between year and flour type. For APC, we see that the lines for the years 2024 and 2025 are parallel and this shows there is no block by flour type interaction. Nevertheless, the rest of the response variables show significant block by flour type interaction effect.

4.1 Distribution of E. coli Load by Factors

Boxplots can help us visualize the overall distribution of each of the response variables.

5 Interpretation of boxplots

The boxplots above show the first, second and third quartiles for each response variable, where the second quartile is the mean or median across the years 2024 and 2025. For APC, we see that the means and variances, shown by the middle line and size of boxplots respectively, are not the same. Flour type, BWG has the lowest average APC but with the highest variance. We also see a few outliers in fine flour type and whole grain flour type respectively.

For Coliform, we see that there are a lot of outliers in BWG. Also, we see that there is high variability in the fine flour type and whole grain flour type. From the boxplot, we cannot really tell whether there is a significant difference between fine flour type and whole grain flour, but both flour types differ from BWG.

For EB, we see again that the means for the various flour types differ, with BWG again having the lowest mean EB value. The variability in the three flour types are again different. This is shown in the size of the boxplots. We see similar plot for Yeast.

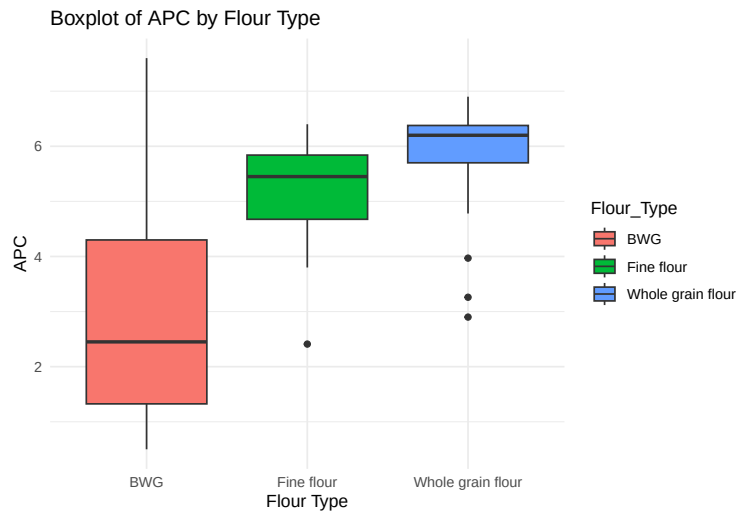


Figure 1: Distribution of *E. coli* Load by Flour Type

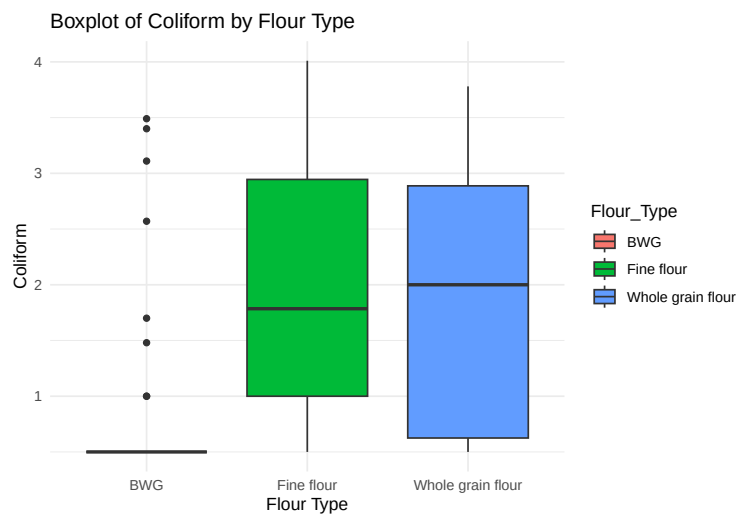


Figure 2: Distribution of *E. coli* Load by Flour Type

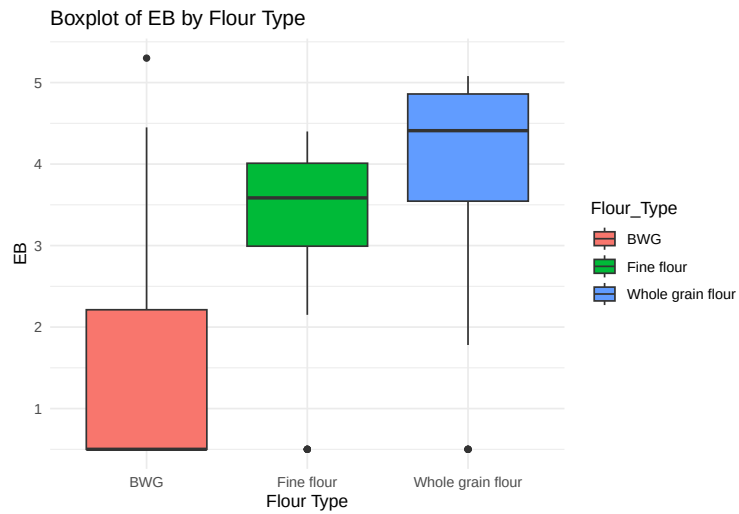


Figure 3: Distribution of *E. coli* Load by Flour Type

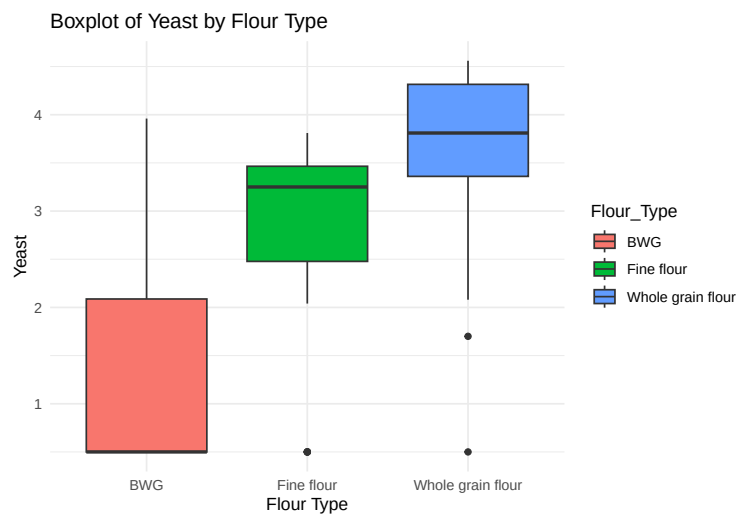


Figure 4: Distribution of *E. coli* Load by Flour Type

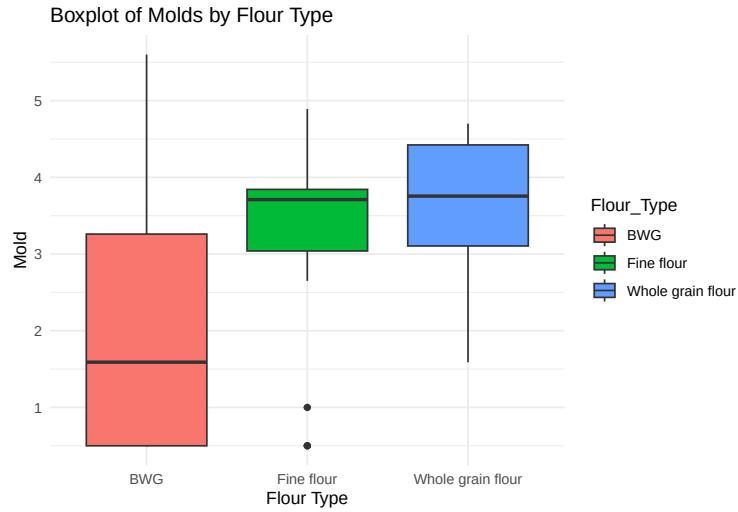
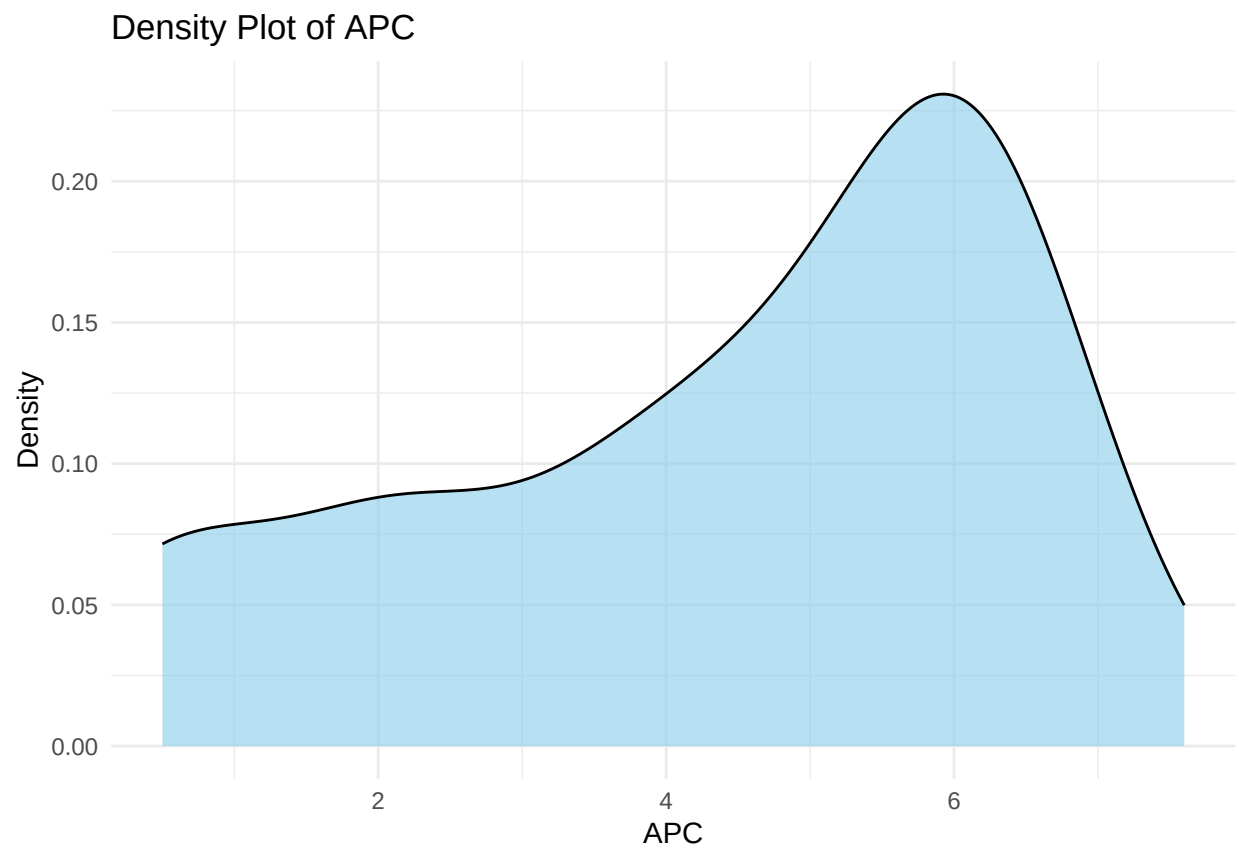
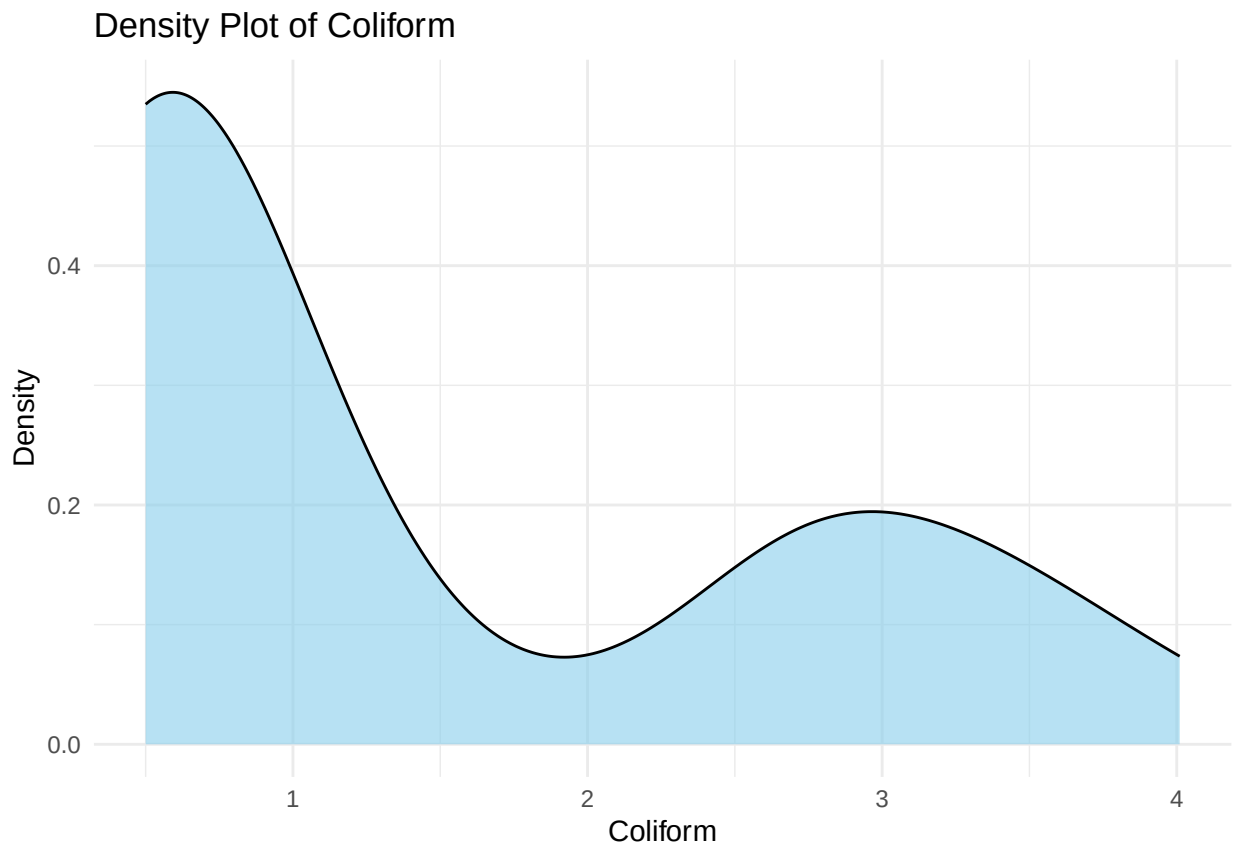


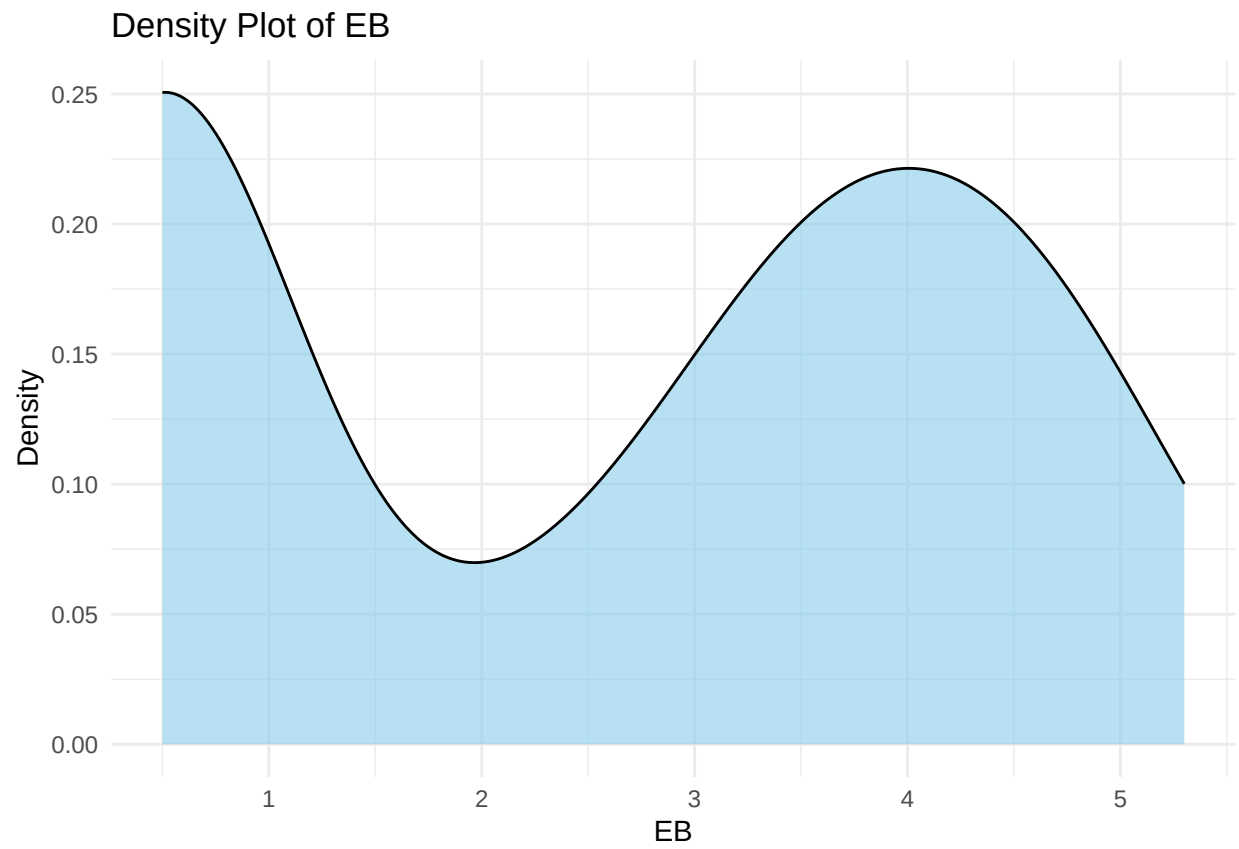
Figure 5: Distribution of *E. coli* Load by Flour Type

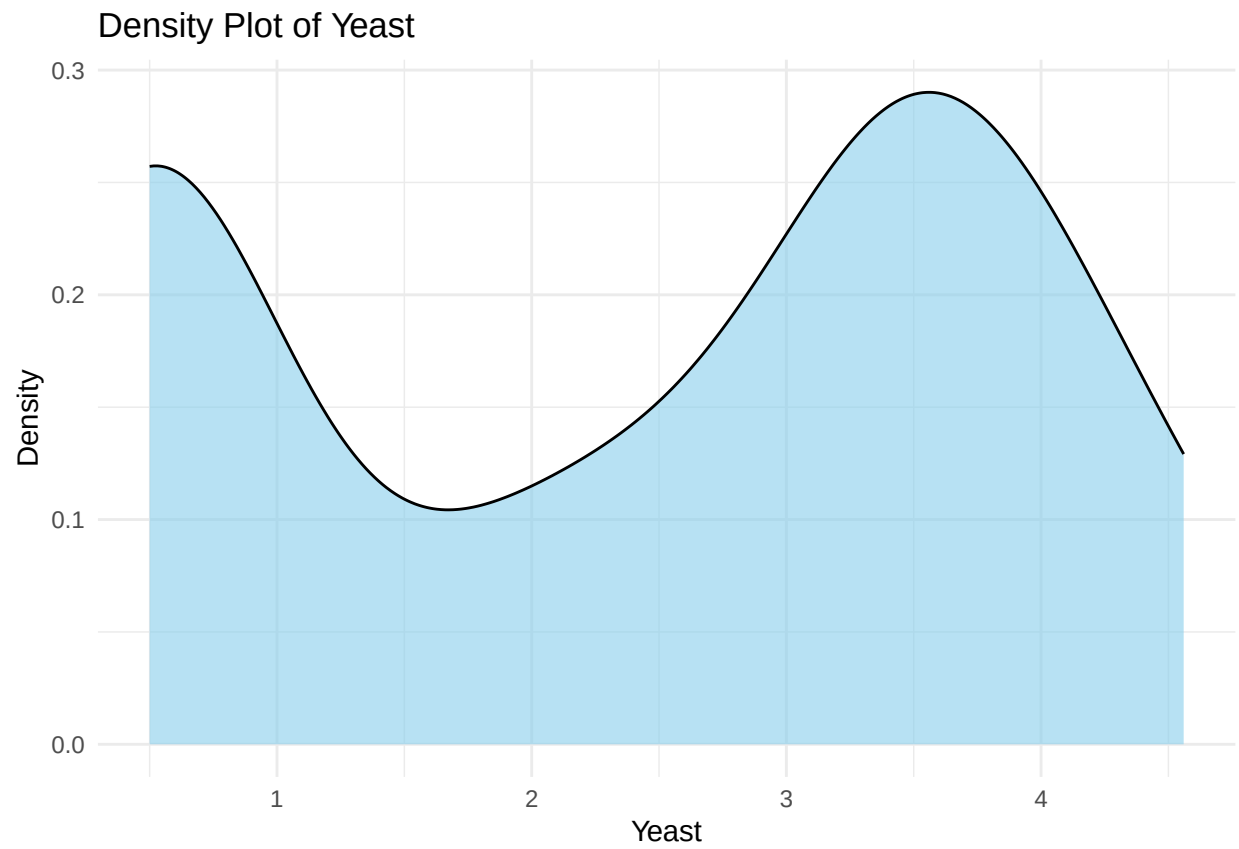
For molds, the mean value for fine flour and whole grain flour types are about the same. This is shown by the middle line in the green and blue boxplots. Nevertheless, whole grain flour has higher variability than fine flour. We also see that BWG has the lowest mean value but the largest variability. Looking at the boxplots for all the 5 response variables tells us that the distribution of the data does not follow the normal distribution. Because the assumption of homogeneous variance fails.

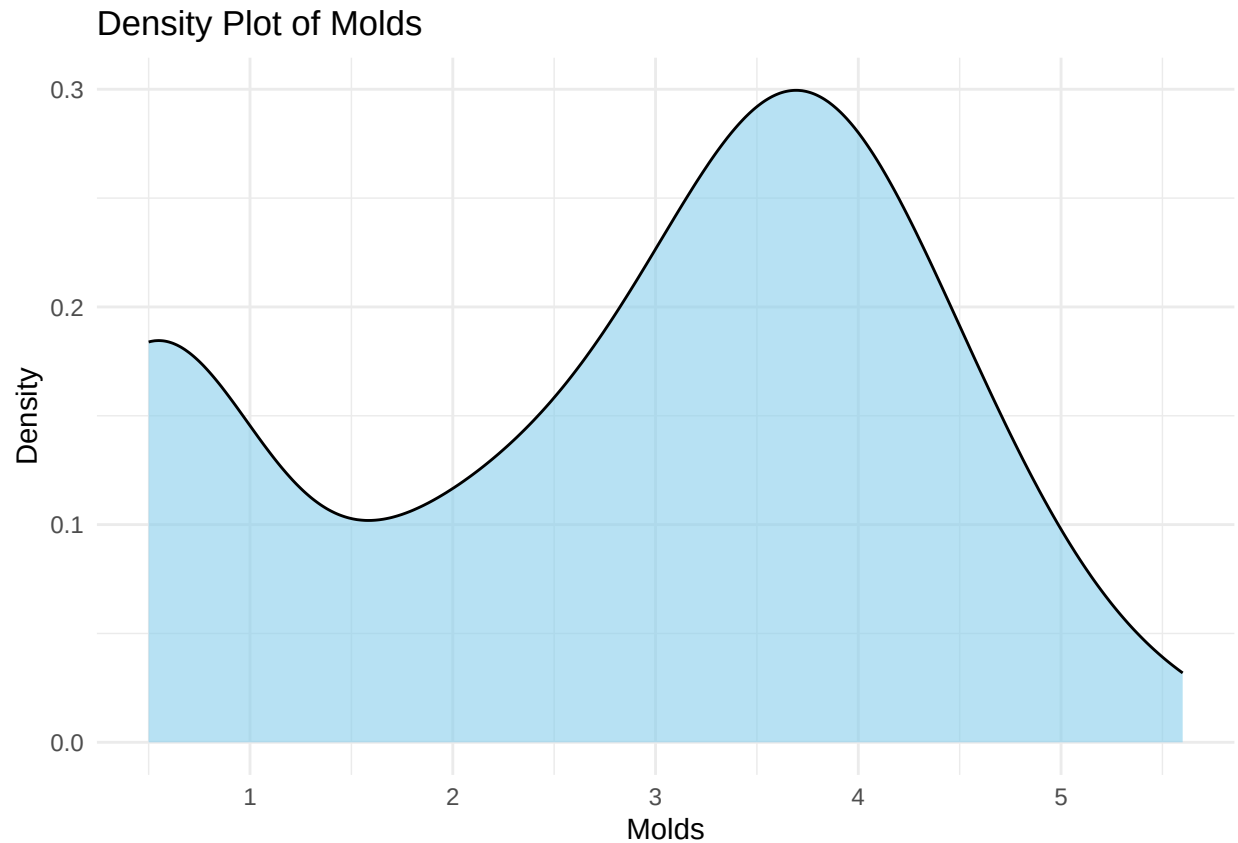
A quick look at the relative density plot shown below confirms that the distribution is not gaussian (normal).





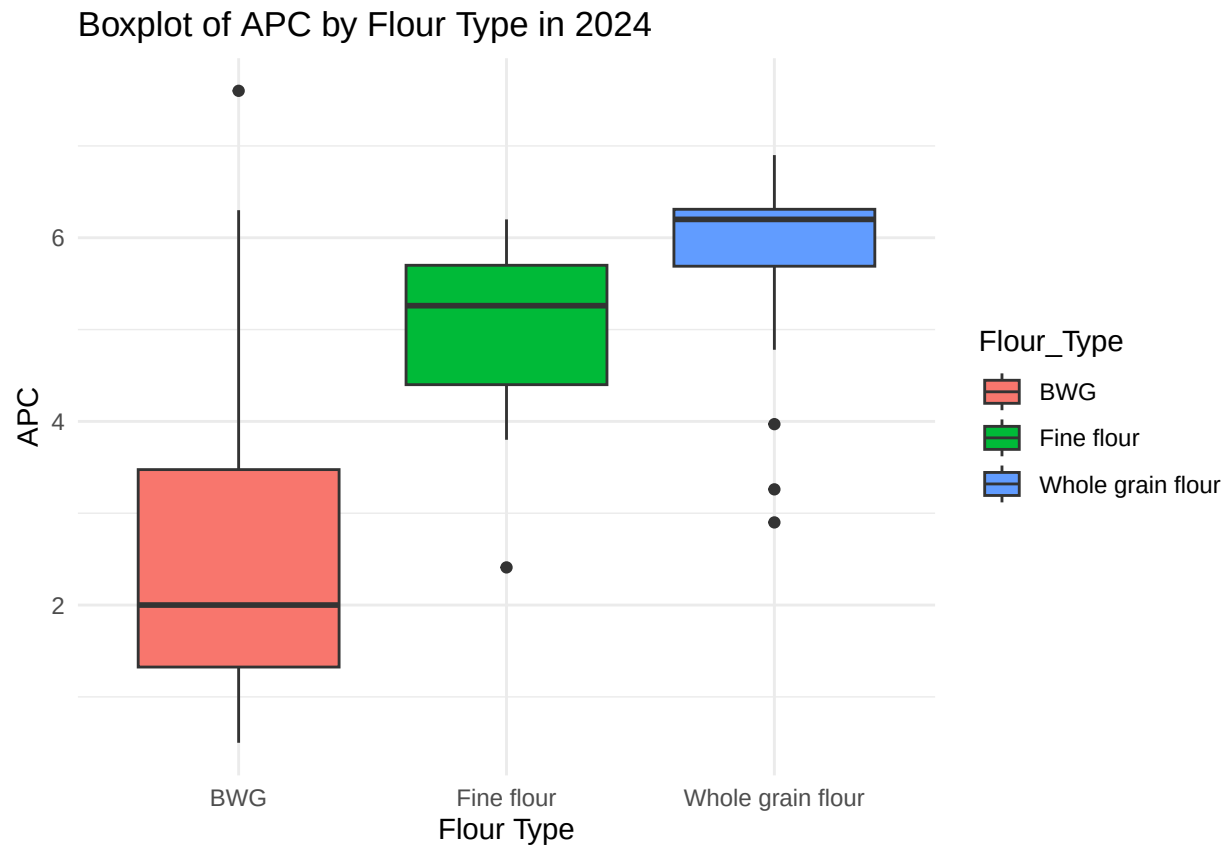


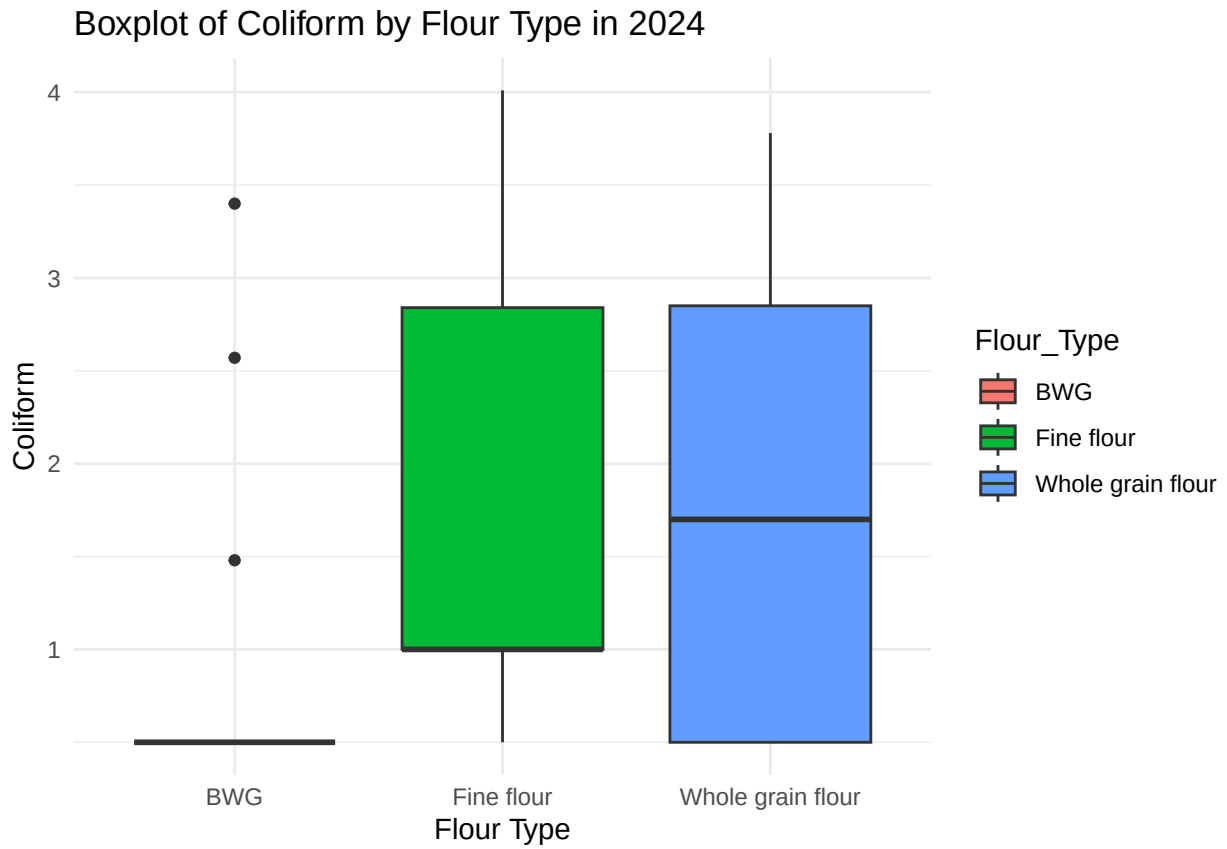


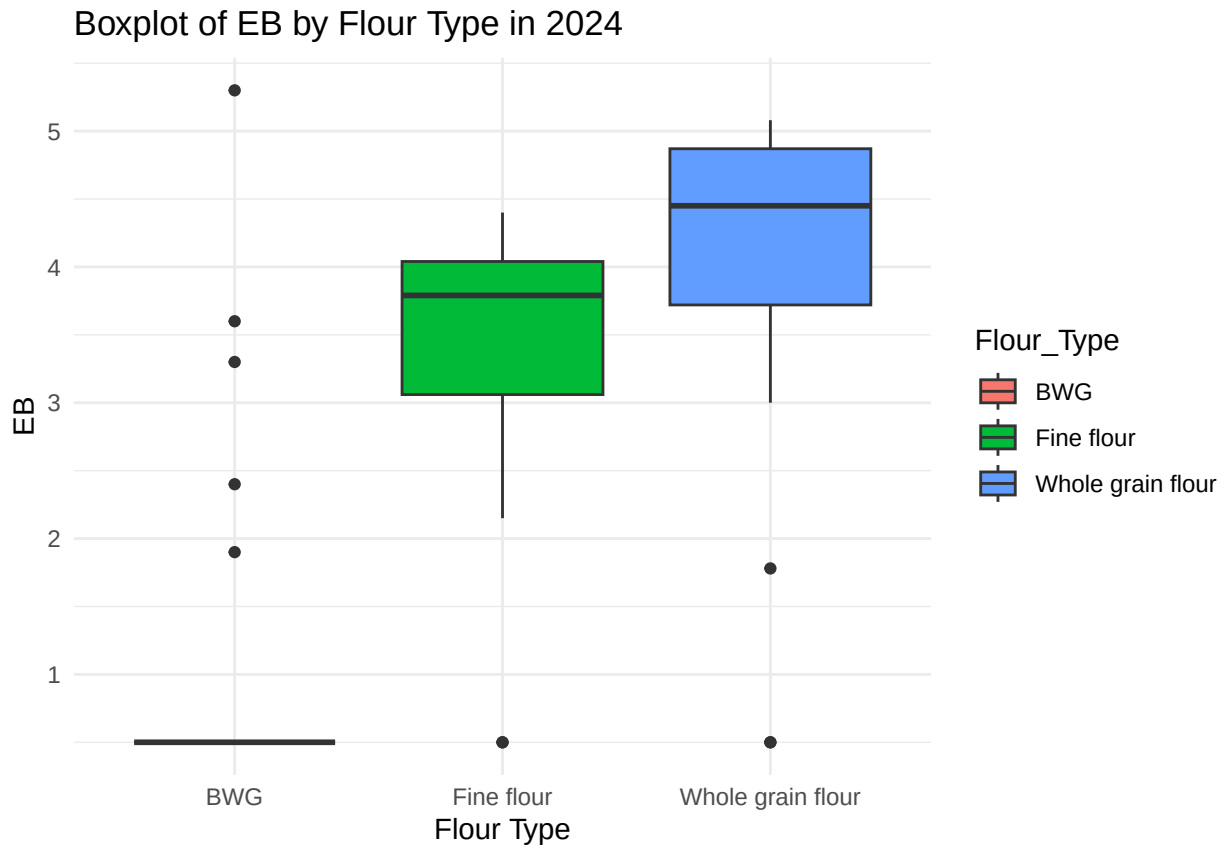


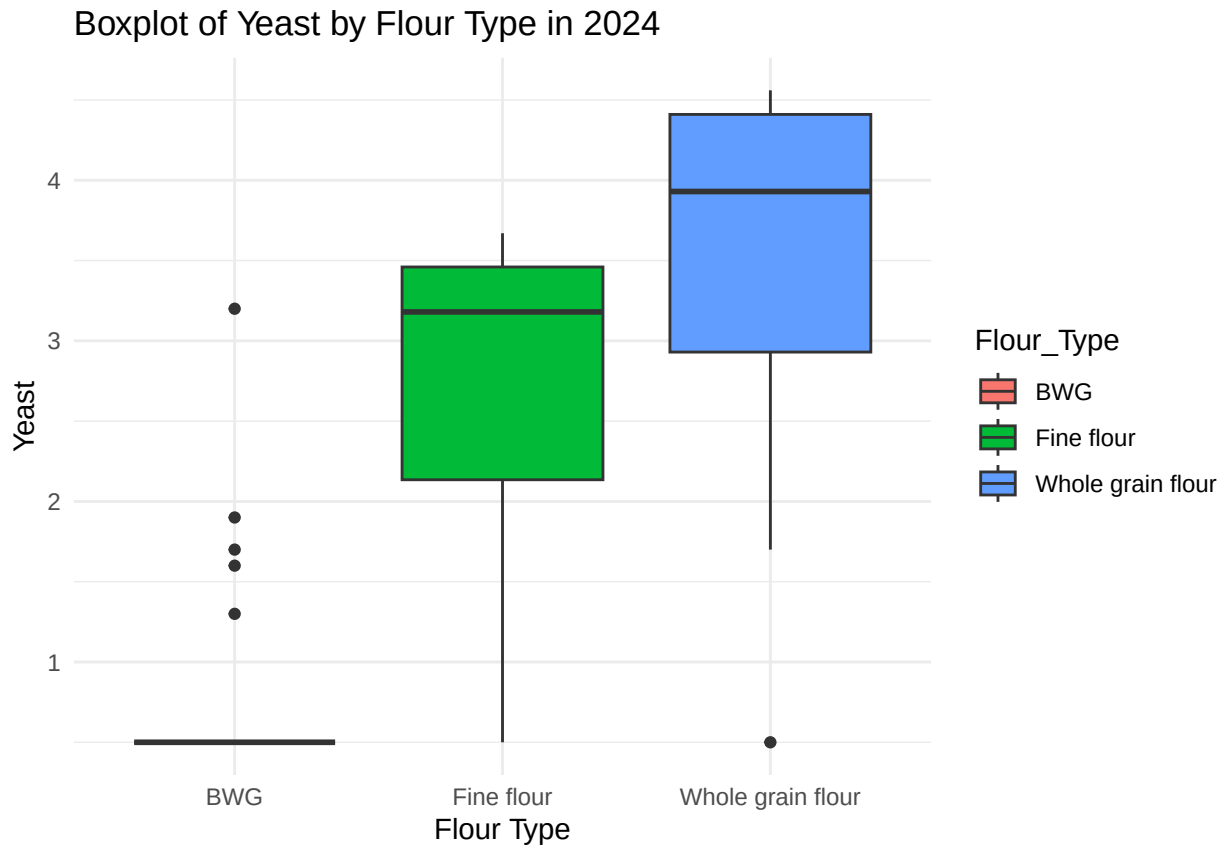
The relative frequency density plots above show that Coliform, EB, Yeast and Molds have bimodal shape. To understand whether the peaks are as a result of flour type differences or the blocks, we subsetting the data and looked at the distribution of all the variables. The distribution of the subsetting data is not shown because the client is not interested in those plots. But from the plots, we still see bimodal shapes for the variables which show that the peaks are as a result of the treatments. This in itself tells us that at least one flour type differs from the others.

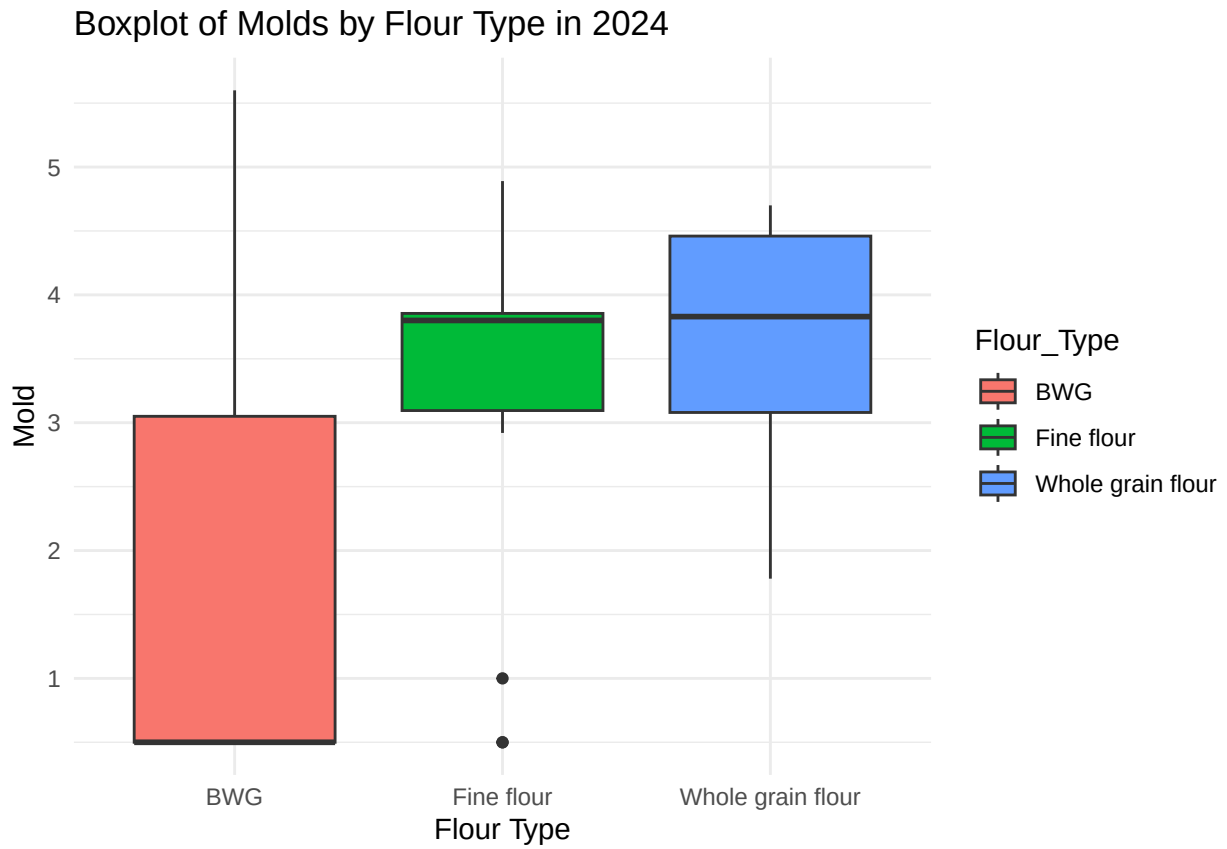
Nevertheless, we show the boxplot for each treatment in each year. The plots below show boxplots for each flour type and for each variable in the year 2024.



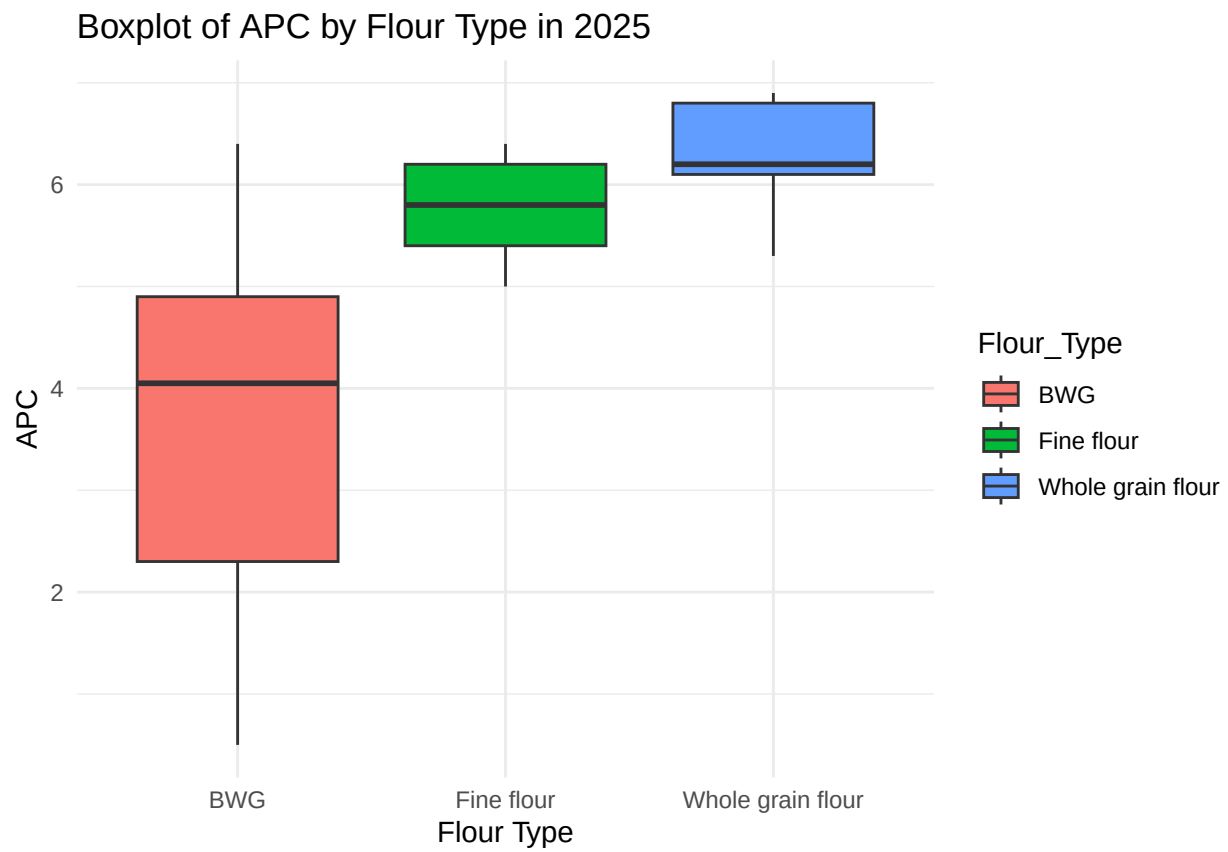


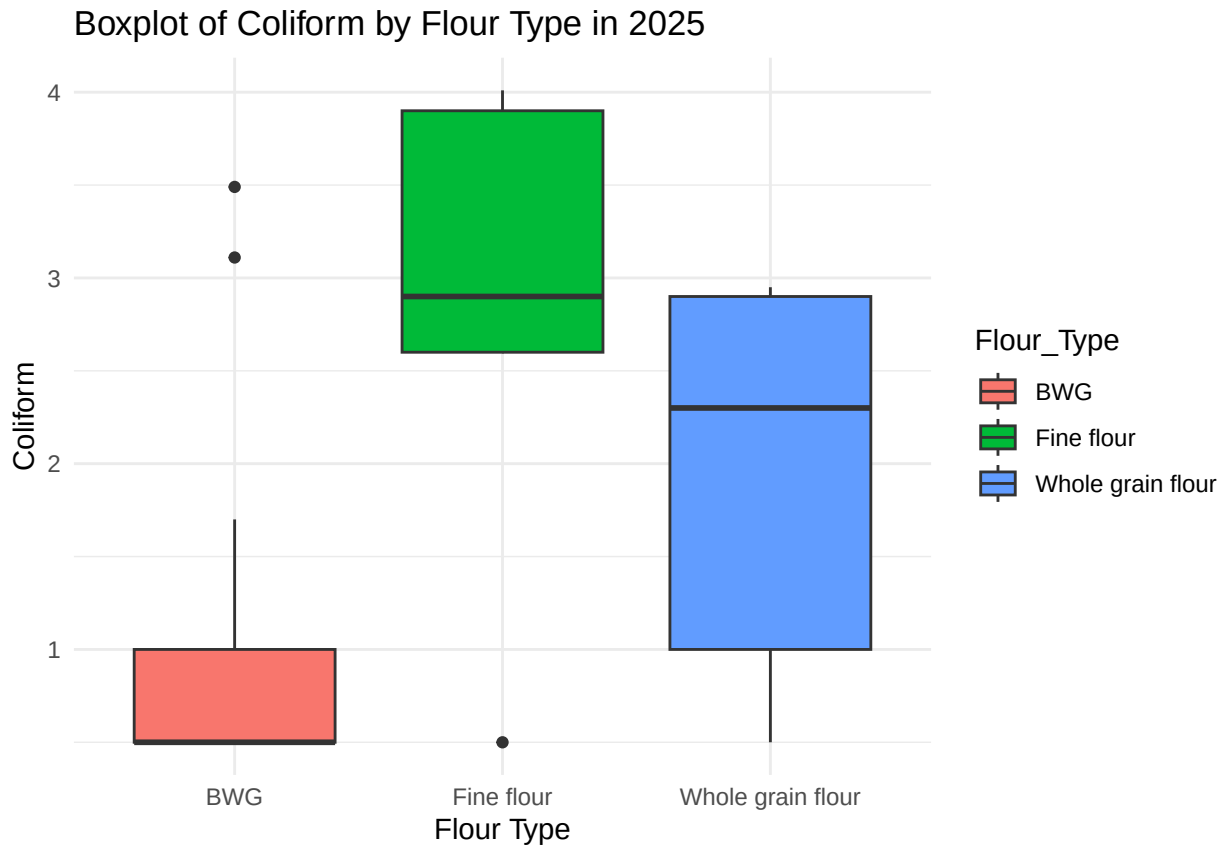


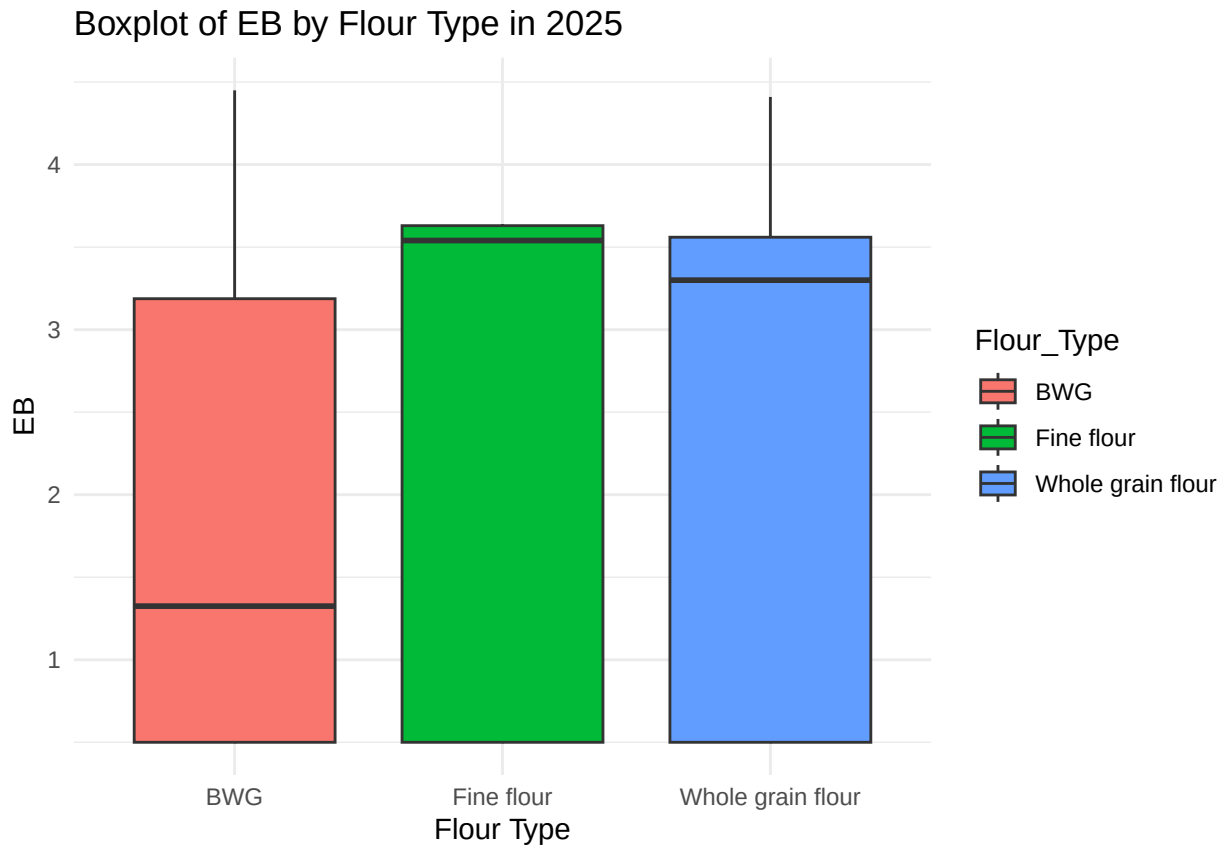


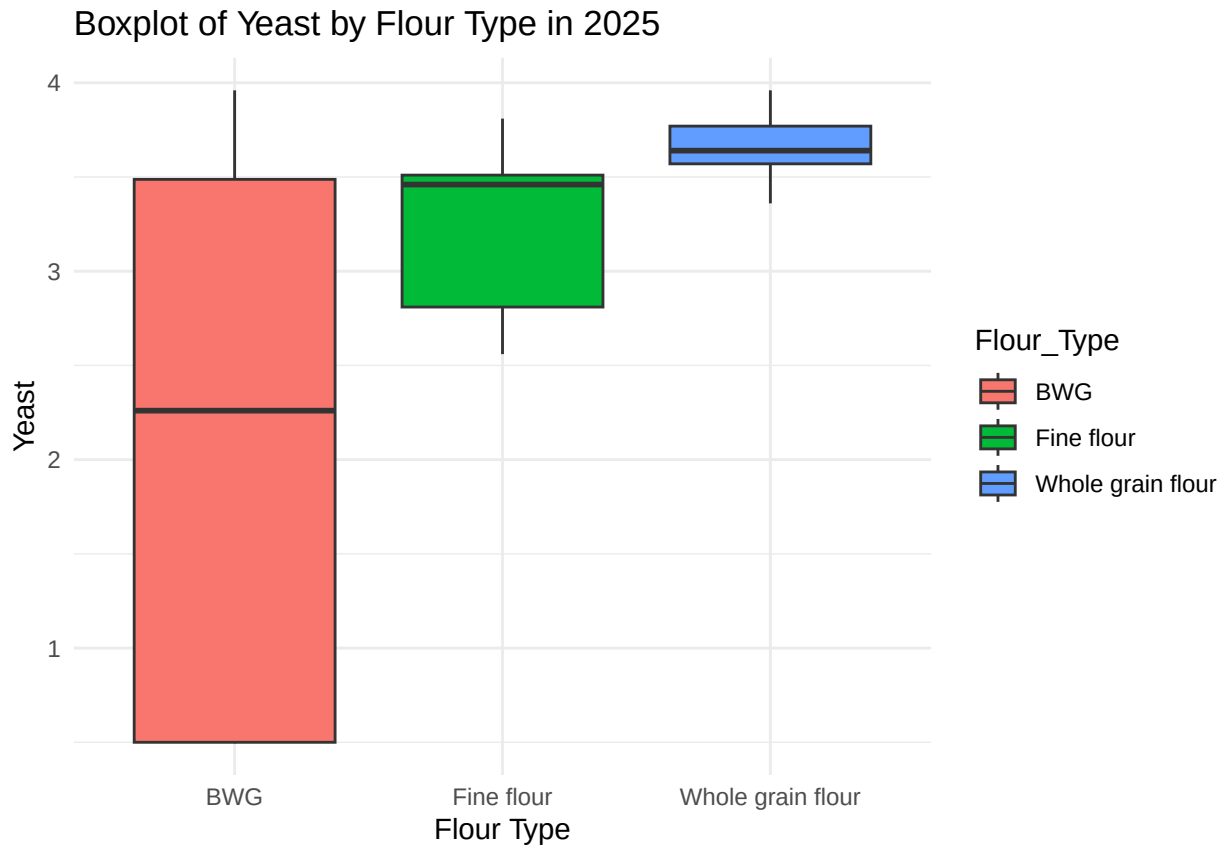


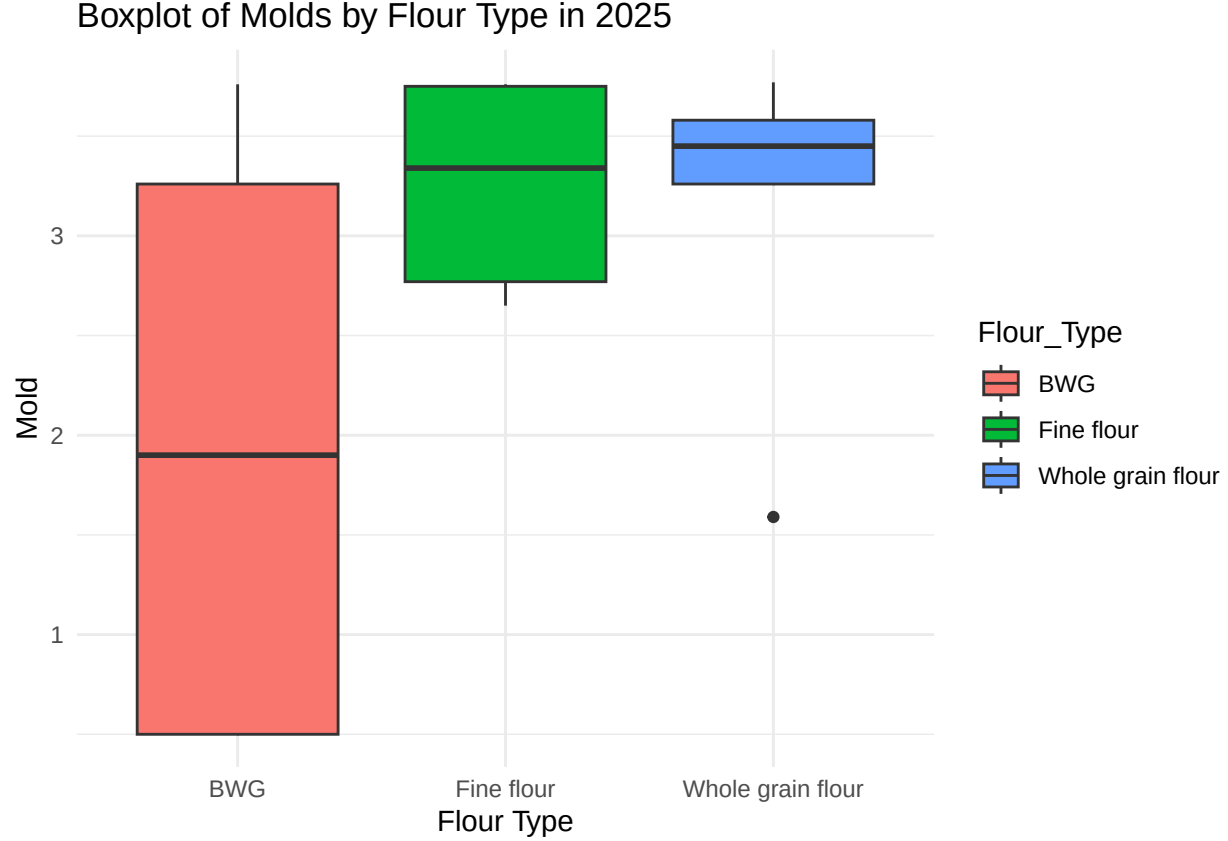
The boxplots below for each flour type in the year 2025 for each variable is shown.











Based on the boxplots, we analyze the data for the two years separately. The rest of the analysis is done in SAS using a generalized linear model. Results from SAS, are show in this file and the SAS code is attached for client's use. From the SAS output, we look at the Fit Statistics table for all the variables and we see that the values for the *PearsonChi-Square/DF* is less than one. This shows that the gamma distribution was a good fit for the data. The plots are not included in the report because the client was not interested in that but code can be run to see the output.

6 Experimental Design and Skeleton ANOVA

The experimental design is a completely randomized design with flour type as treatment and the response variables are assumed to follow a gamma distribution. The general format of the statistical model is given below for all response variables. Note that the same model is used for both data sets.

7 Statistical Model

$$Y_i \sim \text{Gamma}(\mu_i, \phi), \quad i = 1, \dots, n,$$

$$g(\mu_i) = \eta_i = \mu + \text{Flour}_i,$$

where $g(\cdot)$ is the canonical inverse link.

8 Results

The results of the analysis done in SAS is presented below. We show the Fit Statistics tables, Type III Test of Fixed Effects Tables and Differences of Treatment Means tables for flour types in each year.

9 Fit Statistics and Type III Tests of Fixed Effects Tables

Fit Statistics	
-2 Log Likelihood	292.79
AIC (smaller is better)	300.79
AICC (smaller is better)	301.38
BIC (smaller is better)	309.89
CAIC (smaller is better)	313.89
HQIC (smaller is better)	304.41
Pearson Chi-Square	16.53
Pearson Chi-Square / DF	0.24

Type III Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
Flour_Type	2	69	21.29	<.0001

Figure 6: figure 1: Fit Statistics and Type III Test of Fixed Effects Table for APC

For APC, the $\frac{\text{Pearson Chi-Square}}{Df} = 0.24$ which is less than 1 and thus, the model is a good fit. We also show the Type III Tests of Fixed Effects table who shows a significant difference in treatment effect. The $p - \text{value} < 0.0001$ for flour type. The fit statistics for all other variables are presented in the SAS file. Results are shown here and the interpretations are same for all other variables in 2024.

Fit Statistics	
-2 Log Likelihood	155.41
AIC (smaller is better)	163.41
AICC (smaller is better)	164.00
BIC (smaller is better)	172.51
CAIC (smaller is better)	176.51
HQIC (smaller is better)	167.03
Pearson Chi-Square	42.43
Pearson Chi-Square / DF	0.61

Type III Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
Flour_Type	2	69	18.35	<.0001

Figure 7: figure 2: Fit Statistics and Type III Test of Fixed Effects Table for Coliform

Fit Statistics and Type III Test of Fixed Effects Table for Coliform is shown and we see that there is a significant difference in the effect of flour type.

Fit Statistics				
-2 Log Likelihood		262.29		
AIC (smaller is better)		270.29		
AICC (smaller is better)		270.90		
BIC (smaller is better)		279.34		
CAIC (smaller is better)		283.34		
HQIC (smaller is better)		273.89		
Pearson Chi-Square		28.86		
Pearson Chi-Square / DF		0.42		

Type III Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
Flour_Type	2	68	8.35	0.0006

Figure 8: figure 3:Fit Statistics and Type III Test of Fixed Effects Table for Molds

Fit Statistics and Type III Test of Fixed Effects Table for Molds is shown and we see that there is a significant difference in the effect of flour type.

Fit Statistics				
-2 Log Likelihood		233.02		
AIC (smaller is better)		241.02		
AICC (smaller is better)		241.62		
BIC (smaller is better)		250.13		
CAIC (smaller is better)		254.13		
HQIC (smaller is better)		244.65		
Pearson Chi-Square		43.72		
Pearson Chi-Square / DF		0.63		

Type III Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
Flour_Type	2	69	36.34	<.0001

Figure 9: figure 4:Fit Statistics and Type III Test of Fixed Effects Table for EB

10 Differences in flour types for 2024 data

For each variable, the client was interested in determining if there was a significant difference among flour types and if so, which flour type differed from the other. We show the Differences of Flour Type Least Square Means Table for each of the variables in 2024.

We see there is a significant difference in effect between BWG and Fine flour, and BWG and Whole grain

Fit Statistics	
-2 Log Likelihood	183.98
AIC (smaller is better)	191.98
AICC (smaller is better)	192.58
BIC (smaller is better)	201.09
CAIC (smaller is better)	205.09
HQIC (smaller is better)	195.61
Pearson Chi-Square	24.50
Pearson Chi-Square / DF	0.36

Type III Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
Flour_Type	2	69	61.32	<.0001

Figure 10: figure 5:Fit Statistics and Type III Test of Fixed Effects Table for Yeast

Differences of Flour_Type Least Squares Means									
Flour_Type	_Flour_Type	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
BWG	Fine flour	-0.7035	0.1501	69	-4.69	<.0001	0.05	-1.0030	-0.4040
BWG	Whole grain flour	-0.8524	0.1390	69	-6.13	<.0001	0.05	-1.1297	-0.5751
Fine flour	Whole grain flour	-0.1489	0.1537	69	-0.97	0.3363	0.05	-0.4556	0.1578

Figure 11: figure 6: Differences of Flour Type for APC in 2024

flour. The p-values are less than 0.0001. But there no significant difference in effect between Fine flour and Whole grain flour, the p-value is 0.3363.

Differences of Flour_Type Least Squares Means									
Flour_Type	_Flour_Type	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
BWG	Fine flour	-0.9256	0.1940	69	-4.77	<.0001	0.05	-1.3126	-0.5386
BWG	Whole grain flour	-0.9772	0.1796	69	-5.44	<.0001	0.05	-1.3355	-0.6190
Fine flour	Whole grain flour	-0.05165	0.1987	69	-0.26	0.7956	0.05	-0.4480	0.3446

Figure 12: figure 7: Differences of Flour Type for Coliform in 2024

We again see there is a significant difference in effect between BWG and Fine flour, and BWG and Whole grain flour. The p-values are less than 0.0001. But there no significant difference in effect between Fine flour and Whole grain flour, the p-value is 0.7956.

Differences of Flour_Type Least Squares Means									
Flour_Type	_Flour_Type	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
BWG	Fine flour	-0.5932	0.1993	68	-2.98	0.0040	0.05	-0.9909	-0.1955
BWG	Whole grain flour	-0.7080	0.1847	68	-3.83	0.0003	0.05	-1.0766	-0.3394
Fine flour	Whole grain flour	-0.1148	0.2026	68	-0.57	0.5729	0.05	-0.5190	0.2894

Figure 13: figure 8: Differences of Flour Type for Molds in 2024

We again see there is a significant difference in effect between BWG and Fine flour, and BWG and Whole grain flour. The p-values are less than 0.0001. But there no significant difference in effect between Fine flour and Whole grain flour, the p-value is 0.5729.

Differences of Flour_Type Least Squares Means									
Flour_Type	_Flour_Type	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
BWG	Fine flour	-1.1835	0.1899	69	-6.23	<.0001	0.05	-1.5623	-0.8047
BWG	Whole grain flour	-1.3981	0.1758	69	-7.95	<.0001	0.05	-1.7488	-1.0474
Fine flour	Whole grain flour	-0.2146	0.1944	69	-1.10	0.2735	0.05	-0.6025	0.1733

Figure 14: figure 9: Differences of Flour Type for EB in 2024

We again see there is a significant difference in effect between BWG and Fine flour, and BWG and Whole grain flour when EB was used as the response. The p-values are less than 0.0001. But there no significant difference in effect between Fine flour and Whole grain flour, the p-value is 0.2735.

Lastly, we see there is a significant difference in effect between BWG and Fine flour, and BWG and Whole grain flour. The p-values are less than 0.0001. But there no significant difference in effect between Fine flour and Whole grain flour, the p-value is 0.0650. But one may consider this p-value as marginally significant.

Similar interpretation is given for results in the 2025 data. We only show the Differences of Flour Type Least Square Means Table for data in 2025.

Differences of Flour_Type Least Squares Means									
Flour_Type	_Flour_Type	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
BWG	Fine flour	-1.2398	0.1592	69	-7.79	<.0001	0.05	-1.5574	-0.9221
BWG	Whole grain flour	-1.5455	0.1474	69	-10.48	<.0001	0.05	-1.8396	-1.2514
Fine flour	Whole grain flour	-0.3057	0.1630	69	-1.88	0.0650	0.05	-0.6310	0.01956

Figure 15: figure 10: Differences of Flour Type for Yeast in 2024

Differences of Flour_Type Least Squares Means									
Flour_Type	_Flour_Type	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
BWG	Fine flour	-0.4665	0.3164	23	-1.47	0.1540	0.05	-1.1212	0.1881
BWG	Whole grain flour	-0.5498	0.3164	23	-1.74	0.0957	0.05	-1.2044	0.1048
Fine flour	Whole grain flour	-0.08324	0.3906	23	-0.21	0.8331	0.05	-0.8913	0.7248

Figure 16: figure 11: Differences of Flour Type for APC in 2025

11 Differences in flour types for 2024 data

For APC 2025 data, we see that all flour types are not significantly different. The p-values are greater than 0.05.

Differences of Flour_Type Least Squares Means									
Flour_Type	_Flour_Type	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
BWG	Fine flour	-1.0357	0.3505	23	-2.96	0.0071	0.05	-1.7608	-0.3107
BWG	Whole grain flour	-0.6701	0.3505	23	-1.91	0.0684	0.05	-1.3951	0.05494
Fine flour	Whole grain flour	0.3657	0.4326	23	0.85	0.4067	0.05	-0.5294	1.2607

Figure 17: figure 12: Differences of Flour Type for Coliform in 2025

For Coliform 2025 data, we see that only BWG and Fine flour are significantly different but all other flour types are not significantly different. The p-values for BWG and Fine flour is 0.0071 but those of BWG and Whole grain flour and Fine flour and Whole grain flour are greater than 0.05.

For Molds 2025 data, we see that all flour types are not significantly different. The p-values are greater than 0.05.

For EB 2025 data, we see that all flour types are not significantly different. The p-values are greater than 0.05.

Lastly, for Yeast 2025 data, we see that all flour types are not significantly different. The p-values are greater than 0.05.

Differences of Flour_Type Least Squares Means									
Flour_Type	_Flour_Type	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
BWG	Fine flour	-0.5121	0.2904	23	-1.76	0.0911	0.05	-1.1128	0.08866
BWG	Whole grain flour	-0.4732	0.2904	23	-1.63	0.1168	0.05	-1.0739	0.1275
Fine flour	Whole grain flour	0.03885	0.3585	23	0.11	0.9146	0.05	-0.7027	0.7804

Figure 18: figure 13: Differences of Flour Type for Molds in 2025

Differences of Flour_Type Least Squares Means									
Flour_Type	_Flour_Type	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
BWG	Fine flour	-0.1965	0.4344	23	-0.45	0.6552	0.05	-1.0951	0.7021
BWG	Whole grain flour	-0.2347	0.4344	23	-0.54	0.5942	0.05	-1.1333	0.6638
Fine flour	Whole grain flour	-0.03821	0.5362	23	-0.07	0.9438	0.05	-1.1474	1.0710

Figure 19: figure 14: Differences of Flour Type for EB in 2025

Differences of Flour_Type Least Squares Means									
Flour_Type	_Flour_Type	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
BWG	Fine flour	-0.3483	0.2976	23	-1.17	0.2538	0.05	-0.9639	0.2673
BWG	Whole grain flour	-0.4733	0.2976	23	-1.59	0.1254	0.05	-1.0889	0.1424
Fine flour	Whole grain flour	-0.1250	0.3674	23	-0.34	0.7368	0.05	-0.8849	0.6350

Figure 20: figure 15: Differences of Flour Type for Yeast in 2025

12 Conclusion

From the results, the effect of the flour types seems to differ between the two years. Also, the sample size of the data in 2025 and very small, but the client is encouraged to assess what happened between the two years. Was there a significant change in weather and environment conditions. Also, further experiment can be conducted to assess whether treatments truly differ from each other and thus, the differences we see in 2024 is not due to environment or the year.